

CONTROLE DE SISTEMAS AMOSTRADOS

Projeto de Controladores Digitais

Planta Sallen-Key-Duino · Sistema de 2a ordem

Controle II — 2026.1

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Objetivos

1

Modelagem

Função de transferência da planta Sallen-Key

2

Resposta em Frequência

Bode, Nyquist e Nichols

3

Proporcional

Controlador P

4

Avanço e Atraso

Compensador de avanço/atraso de fase

5

PI

Proporcional + Integral

6

Avanço + PI

Compensador em cascata

7

PID

Proporcional + Integral + Derivativo

1

Modelagem — Planta de 2a Ordem

Parâmetros

$$k = 0.652$$

ganho DC

$$\xi = 0.0839$$

amortecimento

$$N = 61 \cdot T = 10 \text{ ms}$$

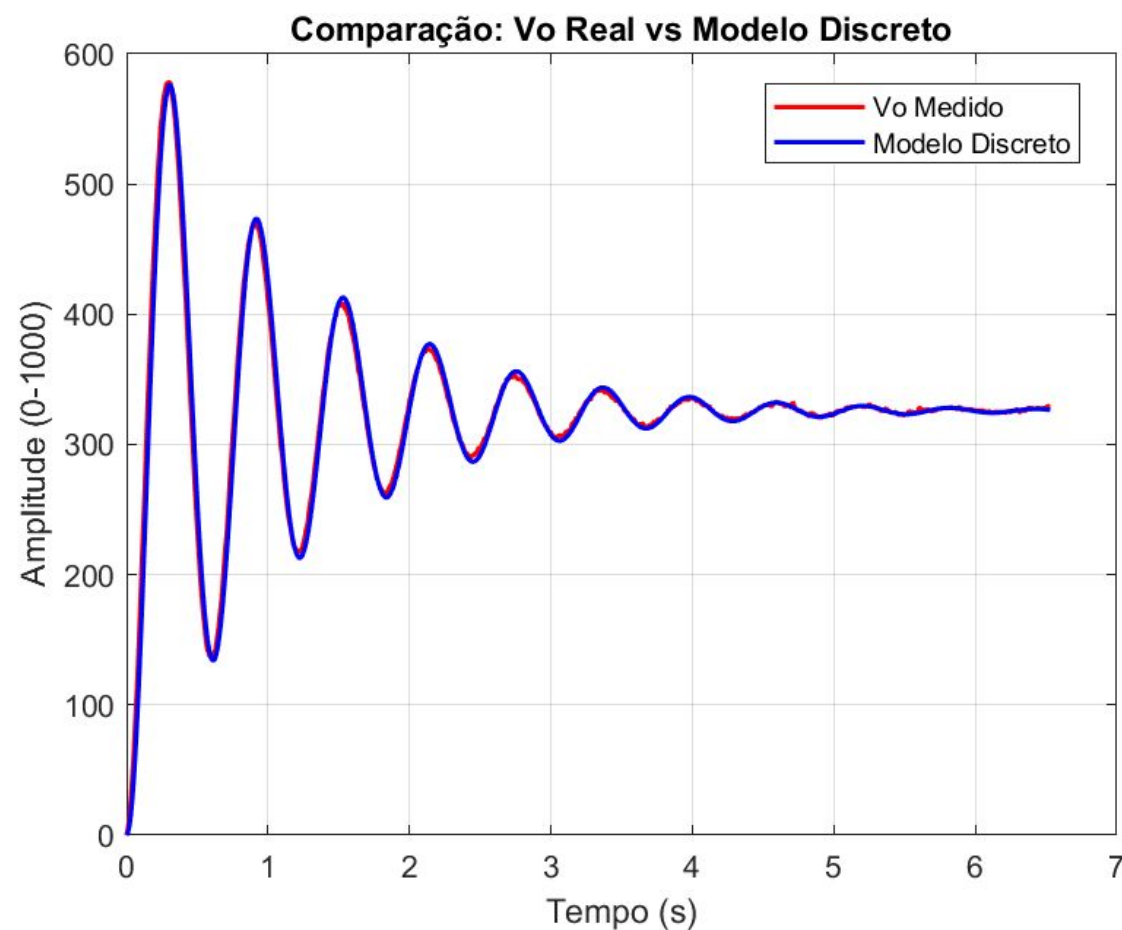
amostras / período

$$W = 10.3$$

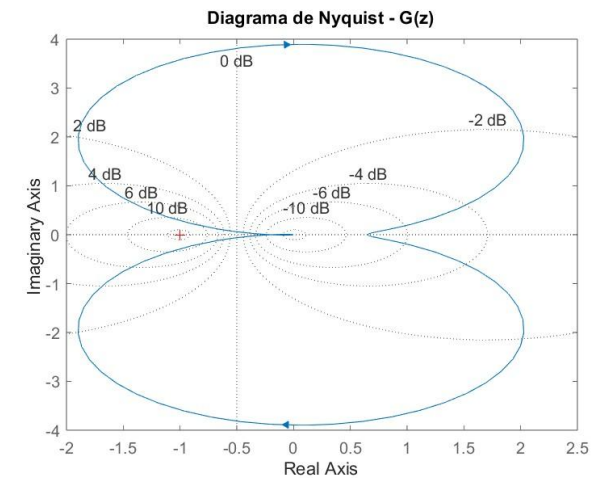
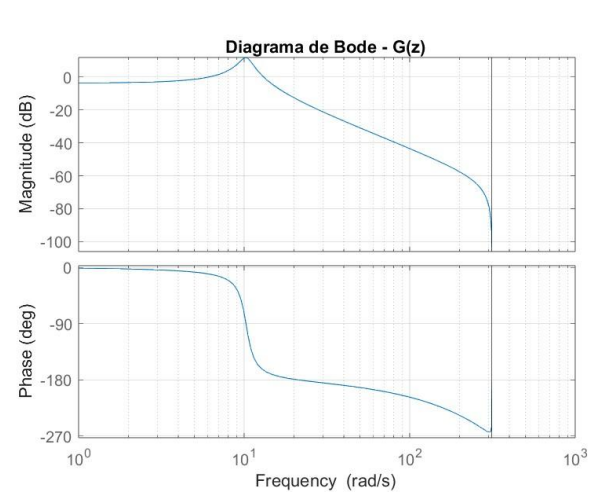
frequência

Discretização: ZOH

$$G(s) := \frac{\omega^2 \cdot k}{s^2 + (2 \cdot \xi \cdot \omega) s + \omega^2} \quad \omega := \frac{2\pi}{n \cdot t} = 10.3 \quad k := \frac{326}{500} = 0.652 \quad M_p := \frac{578 - 327}{327} = 0.768 \quad \zeta := \frac{-\ln(M_p)}{\sqrt{\pi^2 + \ln(M_p)^2}} = 0.084$$



Modelagem — Análises



3 Proporcional

Desempenho

Sobressinal

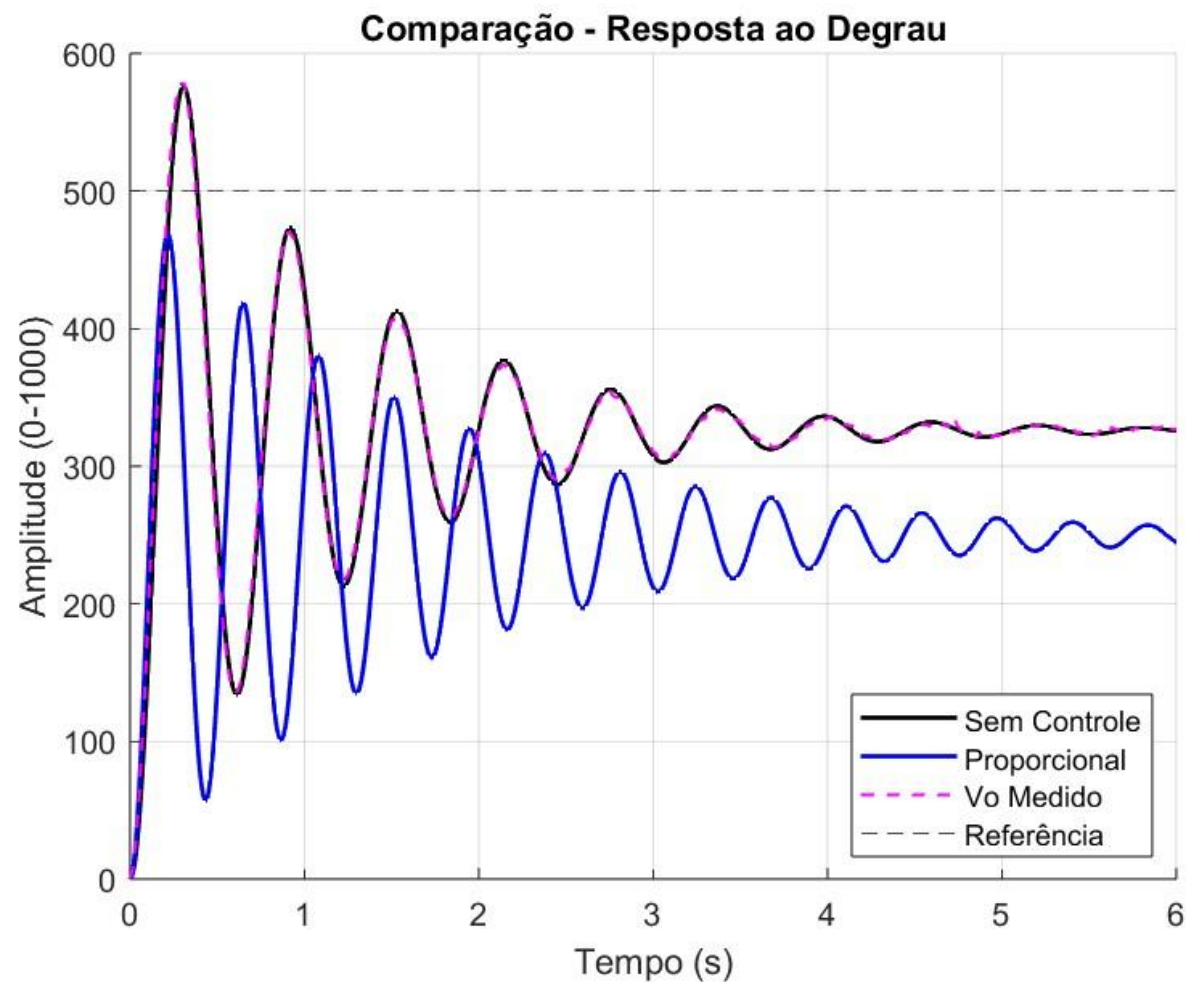
~80 %

Tempo acomod.

<6 s

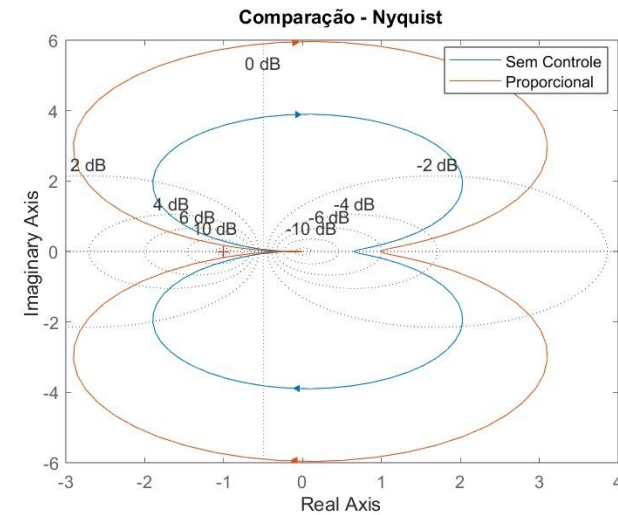
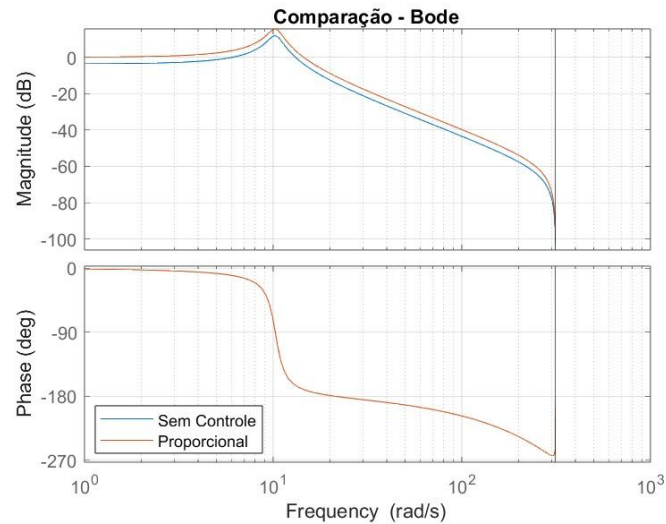
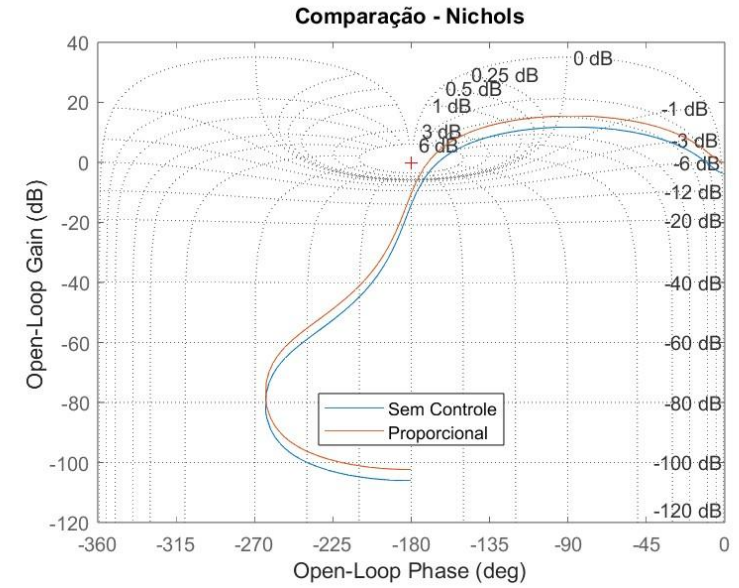
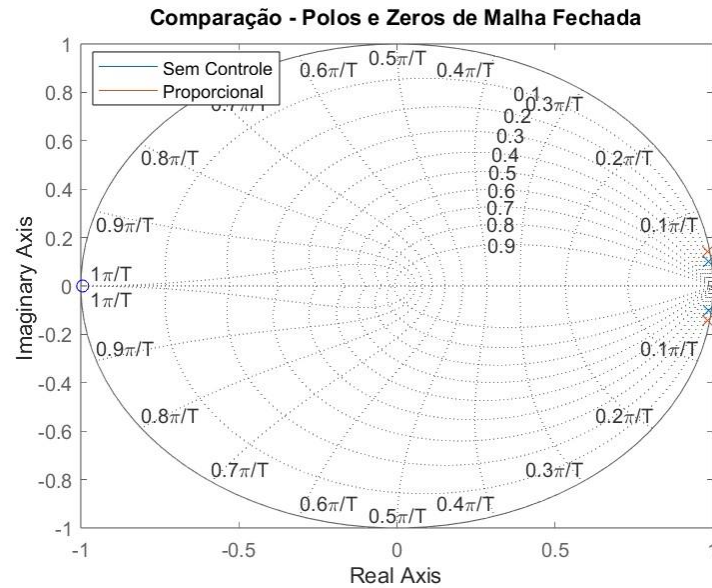
Ganho

$K_p_P = 1.53$



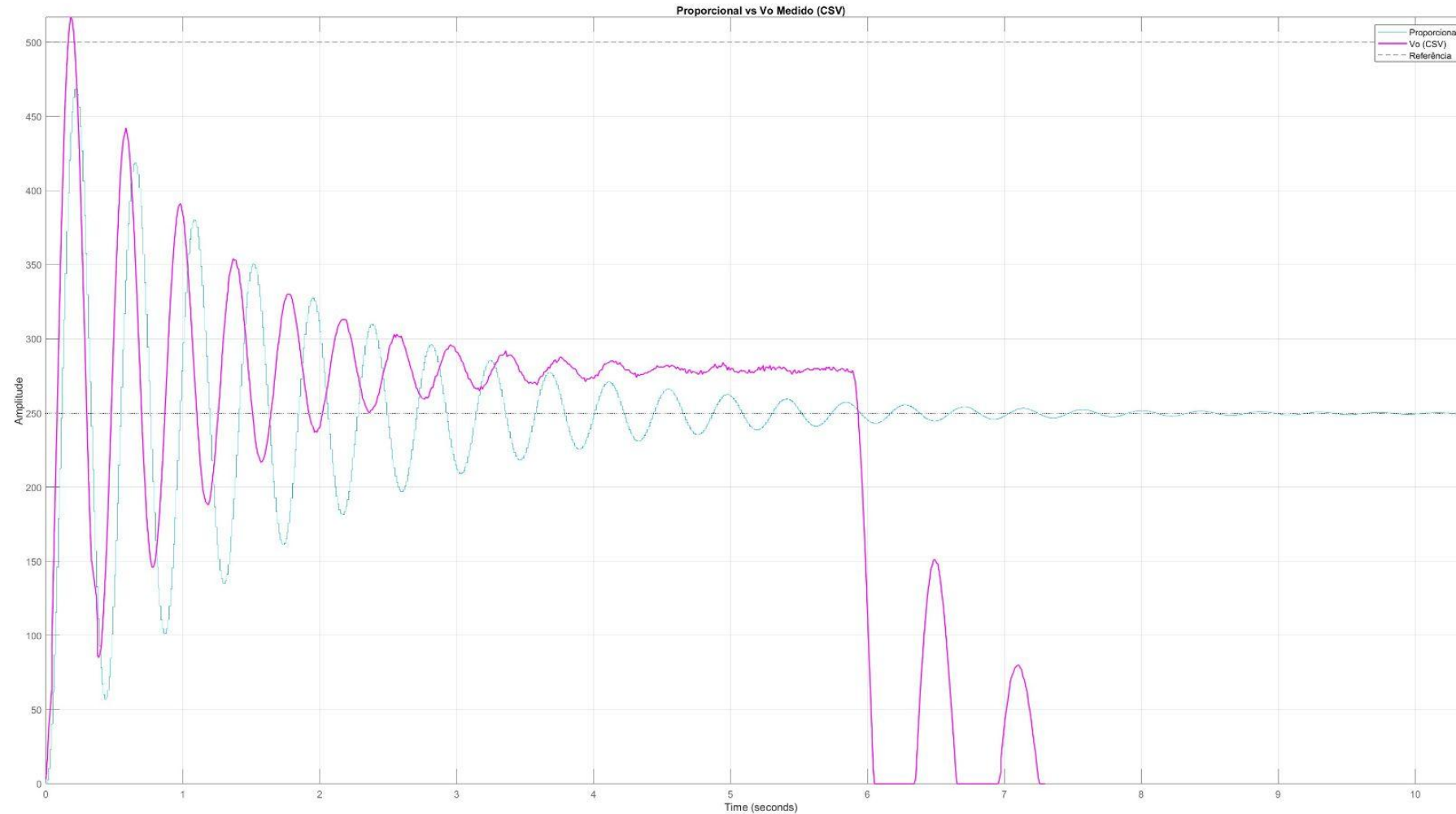
3

Proporcional — Análises



3

Proporcional — (Arduino)



4

Avanço e Atraso

Desempenho

Tipo

Avanço de fase

Zero / Polo

0.85 / -0.85

Ganho Kc

150

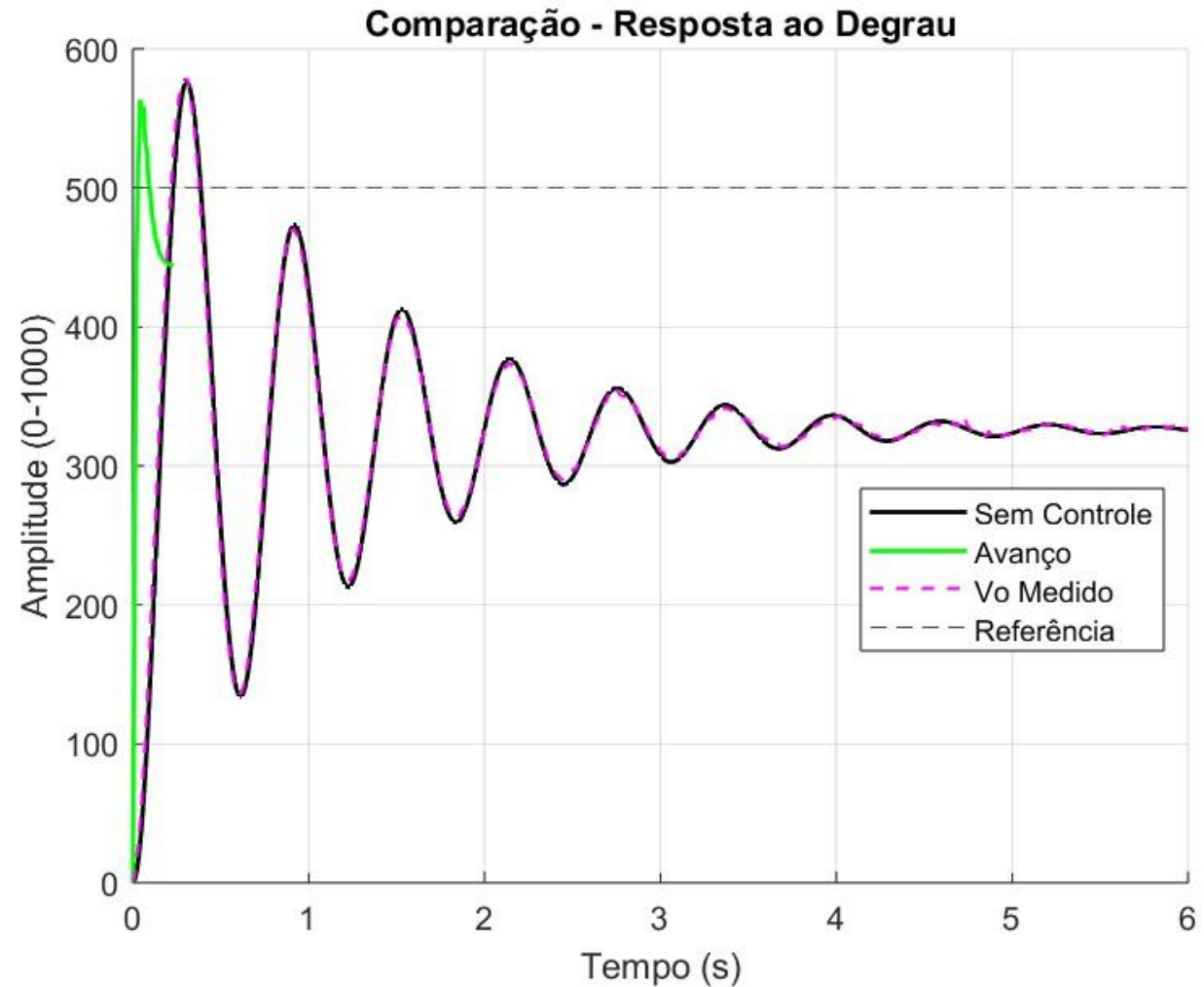
$b_0 = K_c = 150;$

$b_1 = -K_c \cdot z_c = -150 \times 0.85 = -127.5$

$a_1 = -p_c = -(-0.85) = 0.85$

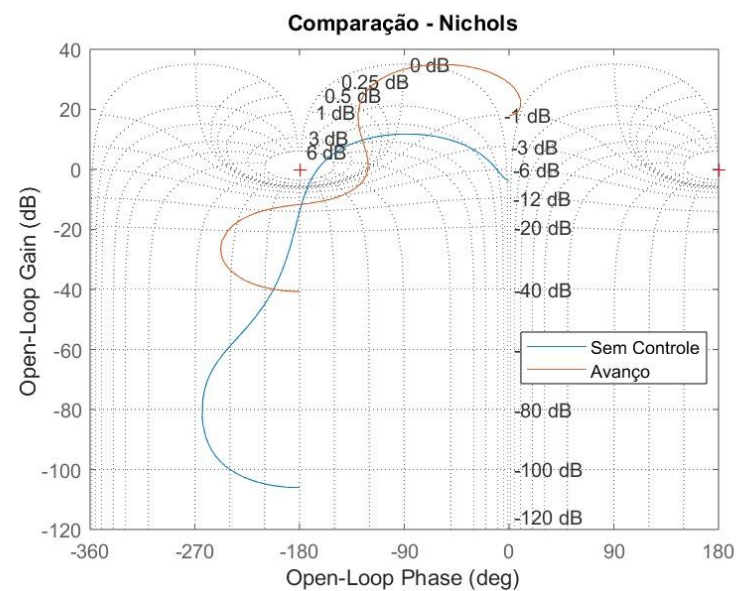
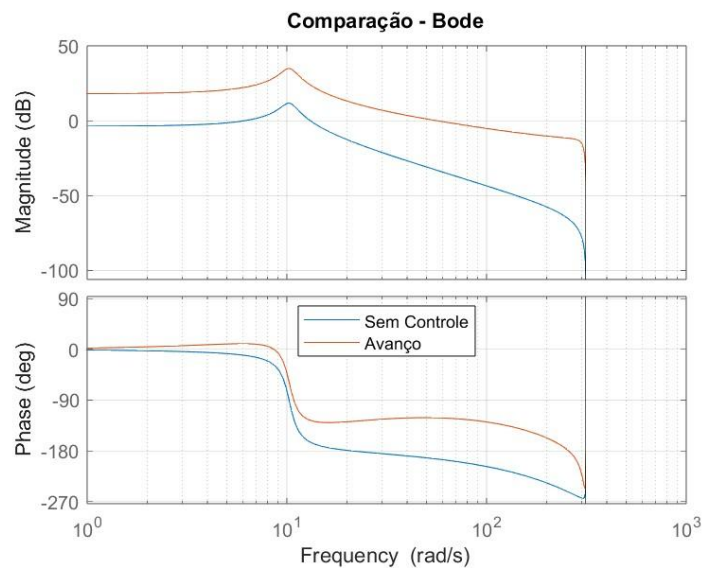
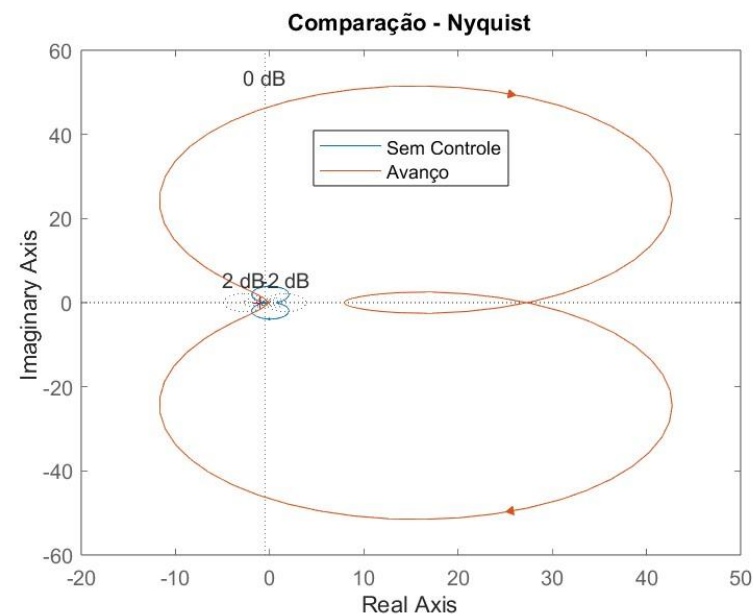
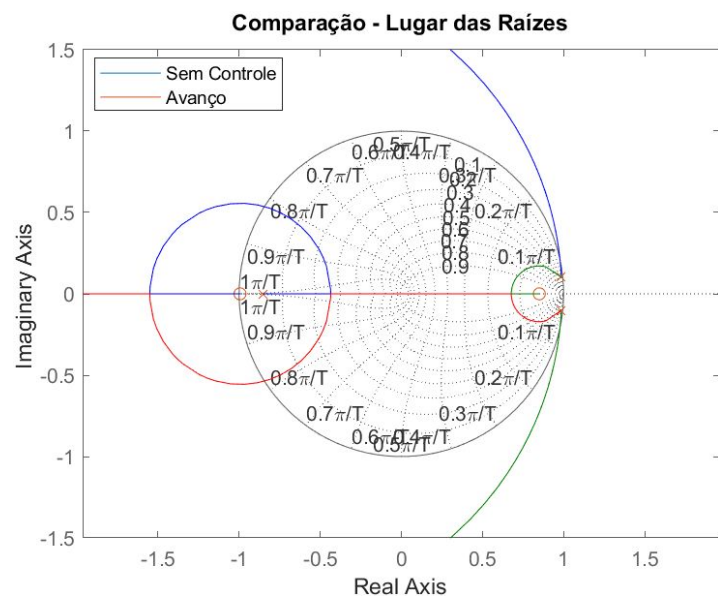
$u[k] = b_0 \cdot e[k] + b_1 \cdot e[k-1] - a_1 \cdot u[k-1]$

$$C(z) := K_c \cdot \frac{(z - Z_c)}{(z - P_v)}$$



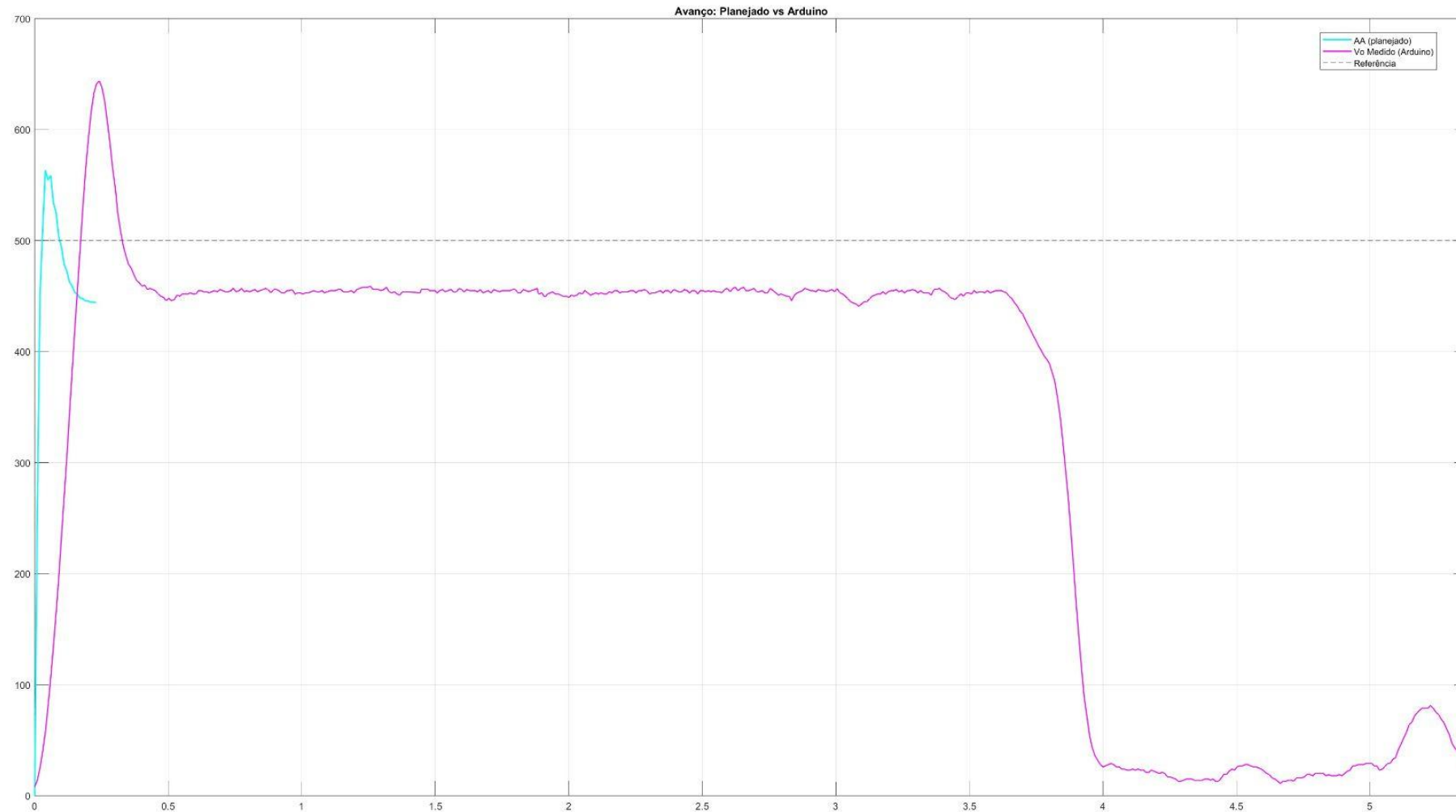
4

Avanço e Atraso — Análises



4

Avanço e Atraso — (Arduino)



5

Controlador P + I

Desempenho

Sobressinal

~1 %

Tempo acomod.

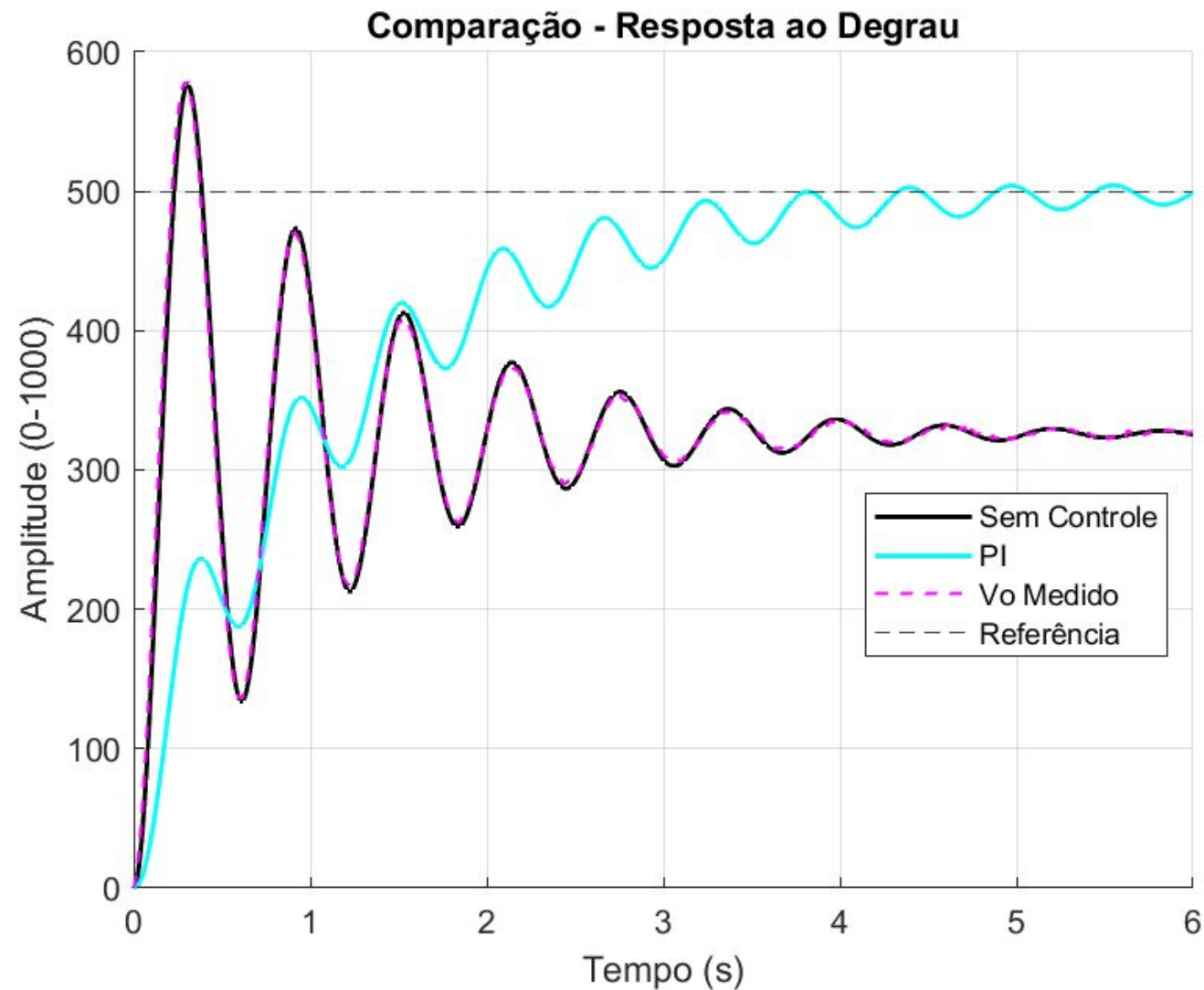
~5 s

Fazendo a conta do Ki:

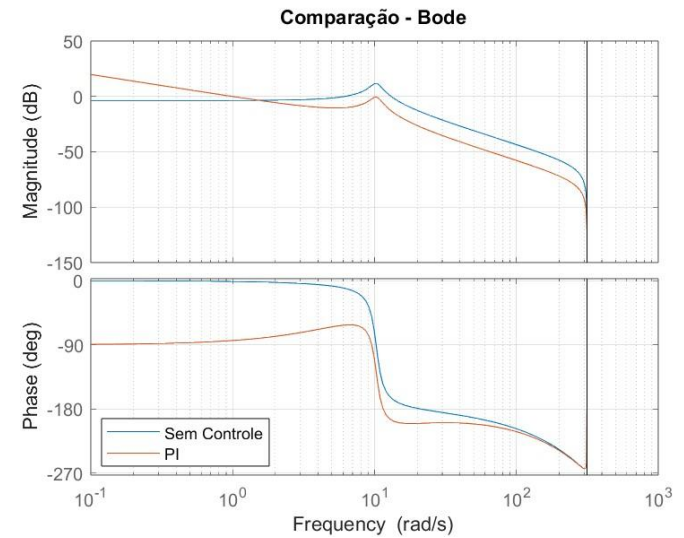
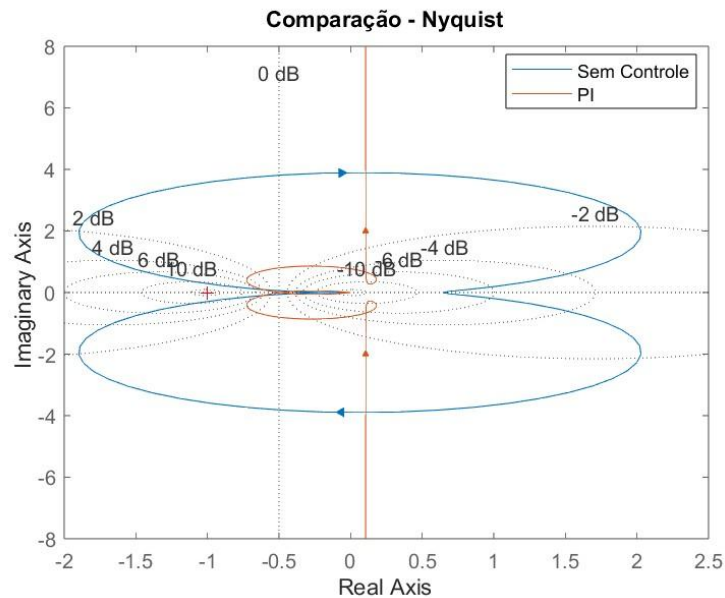
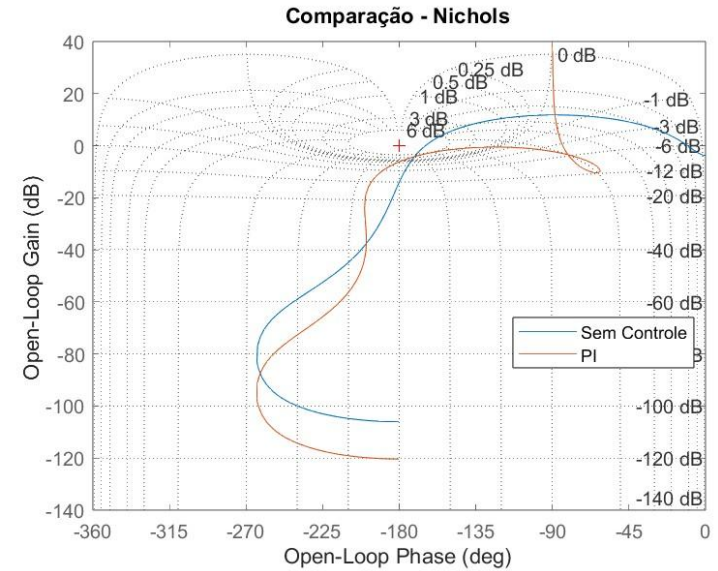
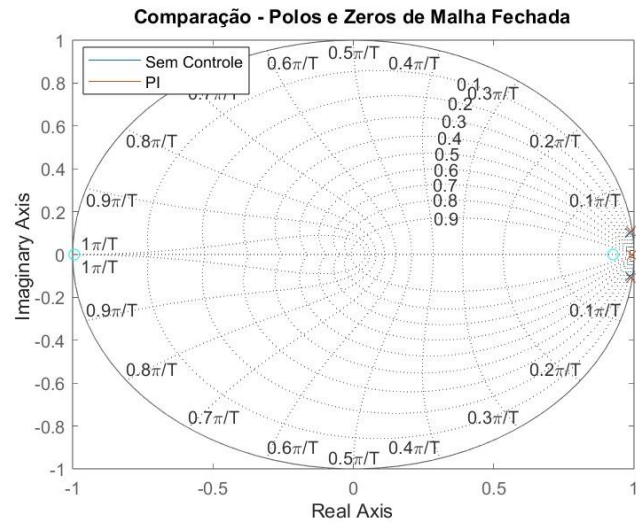
 $z_{c_pi} = 0.925; \quad K_{c_pi} = 0.20;$ $K_{i_PI} = K_{c_pi} \cdot (1 - z_{c_pi}) / T;$ $K_{p_pi} = 0.2; \quad K_{c_pi} = 1.5;$

$$C(z) := K_c \cdot \frac{(z - z_c)}{(Z - 1)}$$

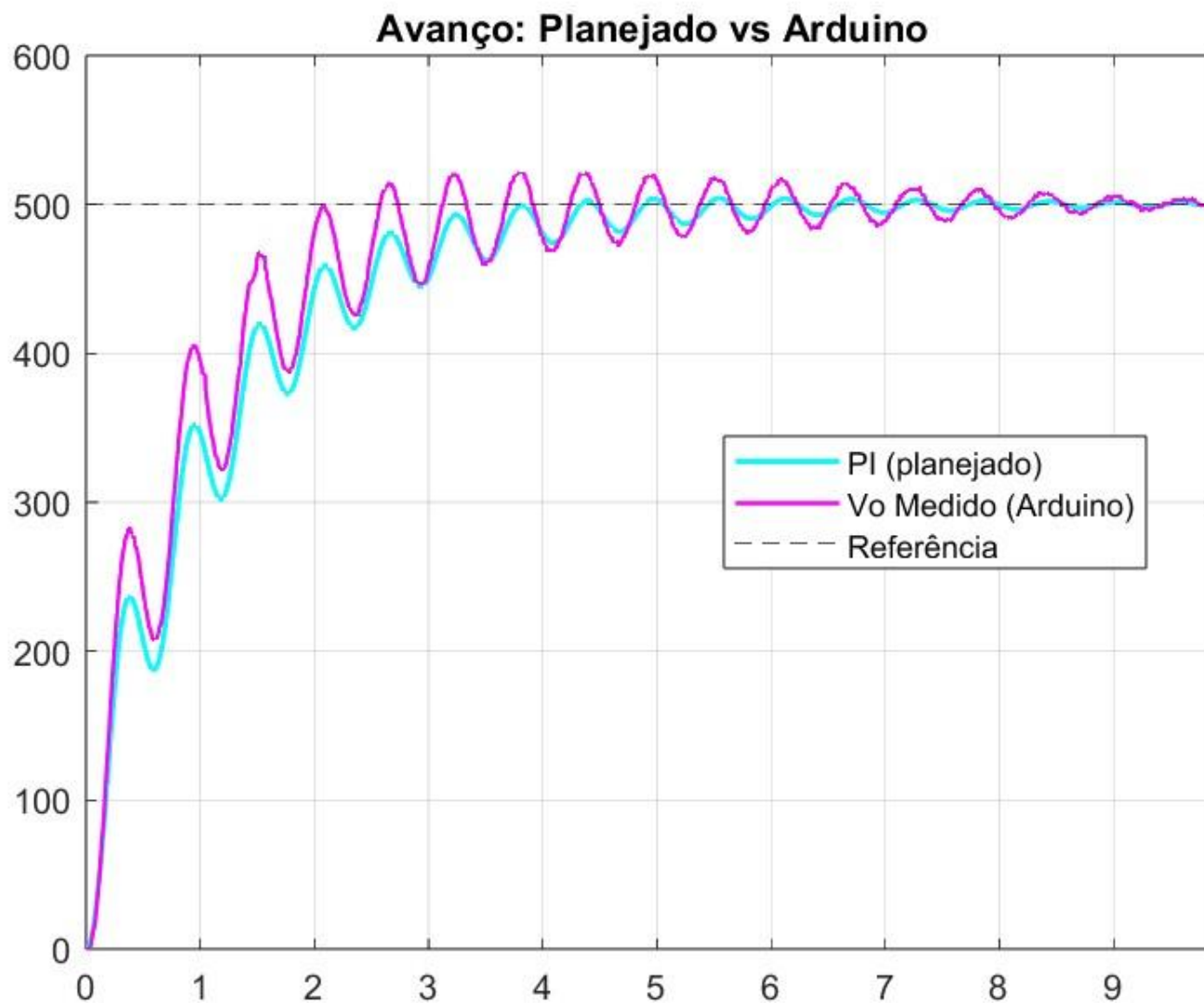
$$C(z) := K_p \cdot \frac{(K_i \cdot T)}{(Z - 1)}$$



Controlador P + I — Análises



Controlador P + I — (Arduino)



6

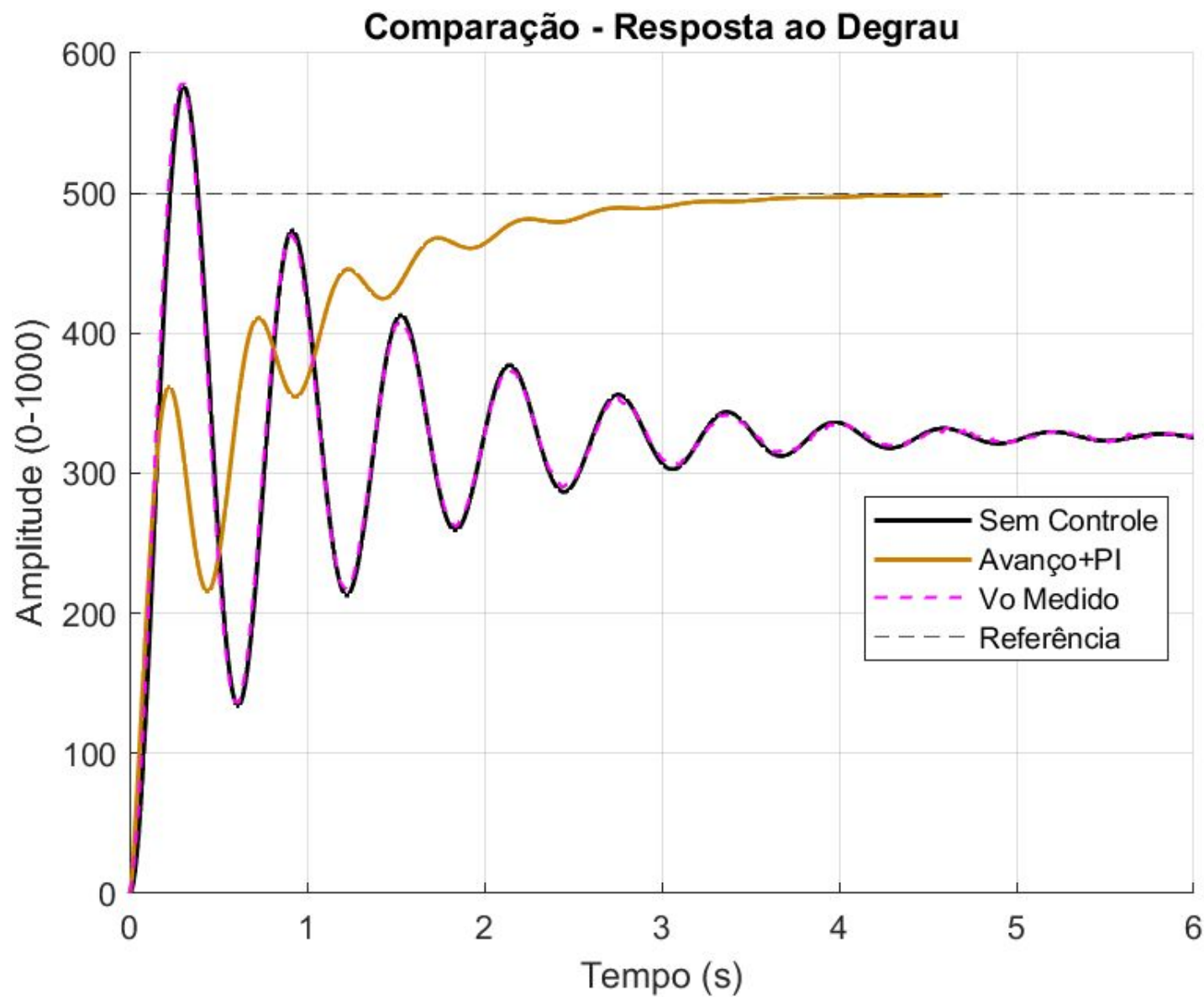
Avanço + PI

Desempenho

Sobressinal

~0 %

Tempo acomod.

~4 s $z_{c_av2} = 0.90; p_{c_av2} = 0.30; K_{c_av2} = 2;$ $z_{c_pi2} = 0.95; K_{c_pi2} = 1.90;$ 

6

Avanço + PI

Parâmetros:

$z_{c_av2} = 0.90$; $pc_av2 = 0.30$; $K_{c_av2} = 2$;

$z_{c_pi2} = 0.95$; $K_{c_pi2} = 1.90$;

Termos intermediários:

Avanço:

$n0 = K_{c_av2} = 2$

$n1 = -K_{c_av2} \cdot z_{c_av2} = -2 \times 0.90 = -1.8$

$d0 = 1$

$d1 = -pc_av2 = -0.30$

PI:

$m0 = K_{c_pi2} = 1.9$

$m1 = -K_{c_pi2} \cdot z_{c_pi2} = -1.9 \times 0.95 = -1.805$

$e0 = 1$

$e1 = -1$

Coefficientes:

$b0 = n0 \cdot m0 = 2 \times 1.9 = 3.800$

$b1 = n0 \cdot m1 + n1 \cdot m0 = 2 \times (-1.805) + (-1.8) \times 1.9 = -7.030$

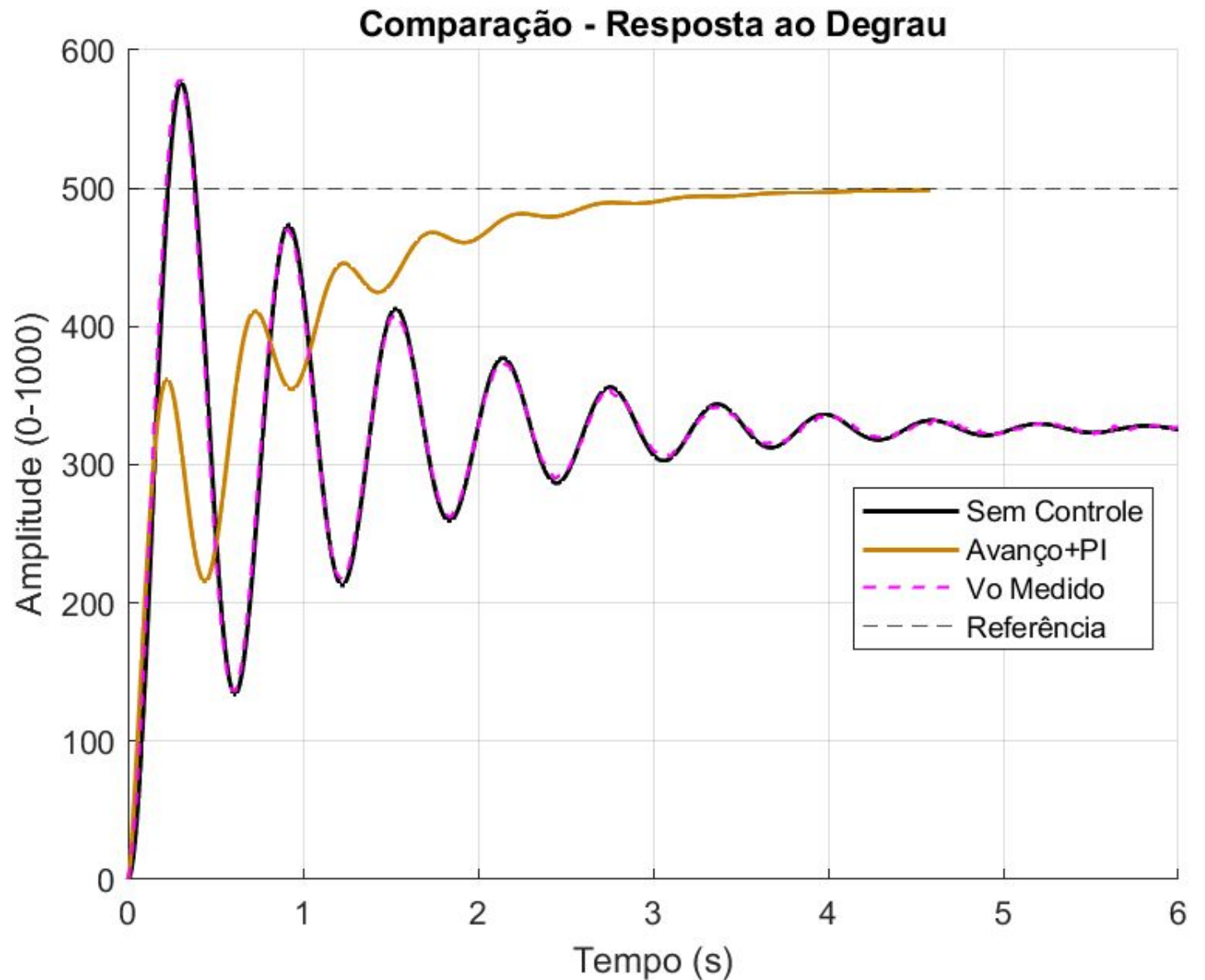
$b2 = n1 \cdot m1 = (-1.8) \times (-1.805) = 3.249$

$a1 = d0 \cdot e1 + d1 \cdot e0 = 1 \times (-1) + (-0.3) \times 1 = -1.300$

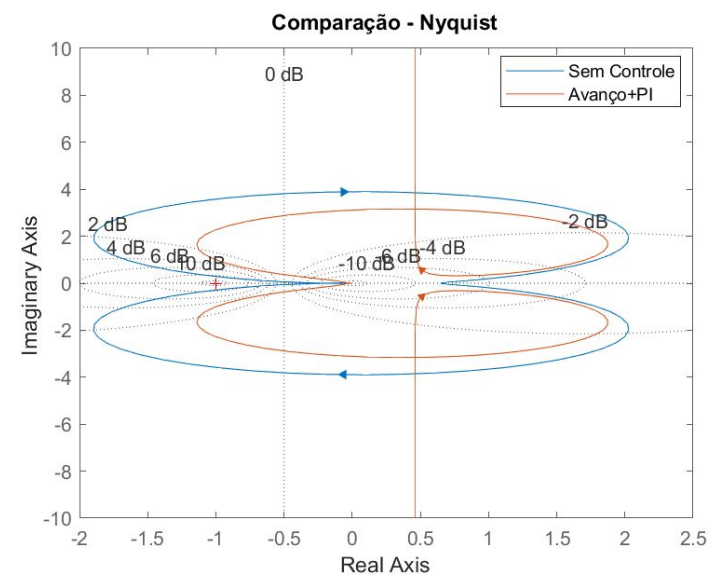
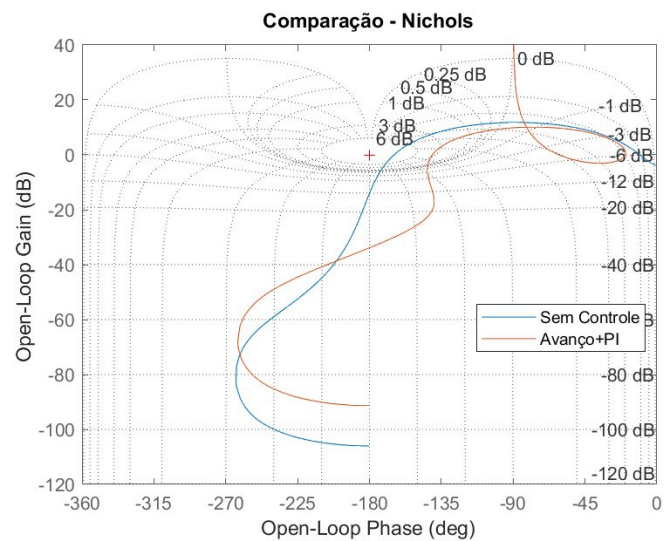
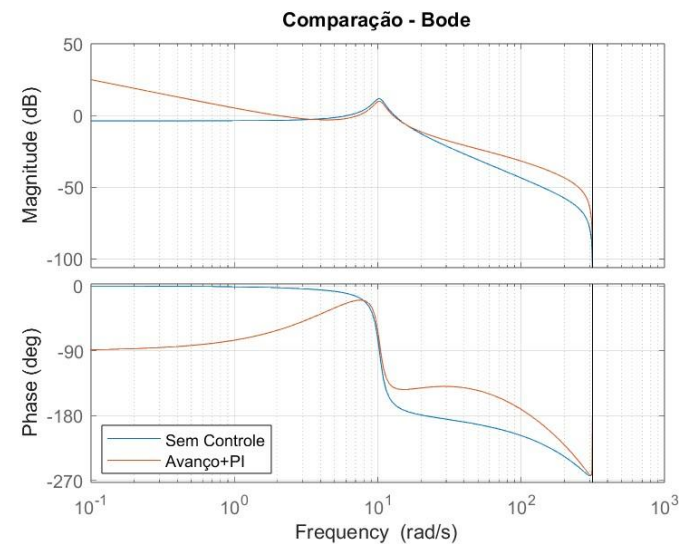
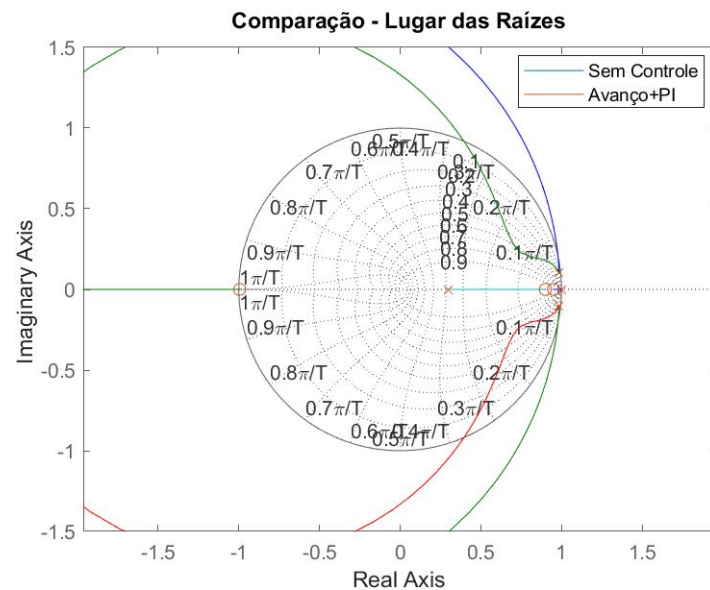
$a2 = d1 \cdot e1 = (-0.3) \times (-1) = 0.300$

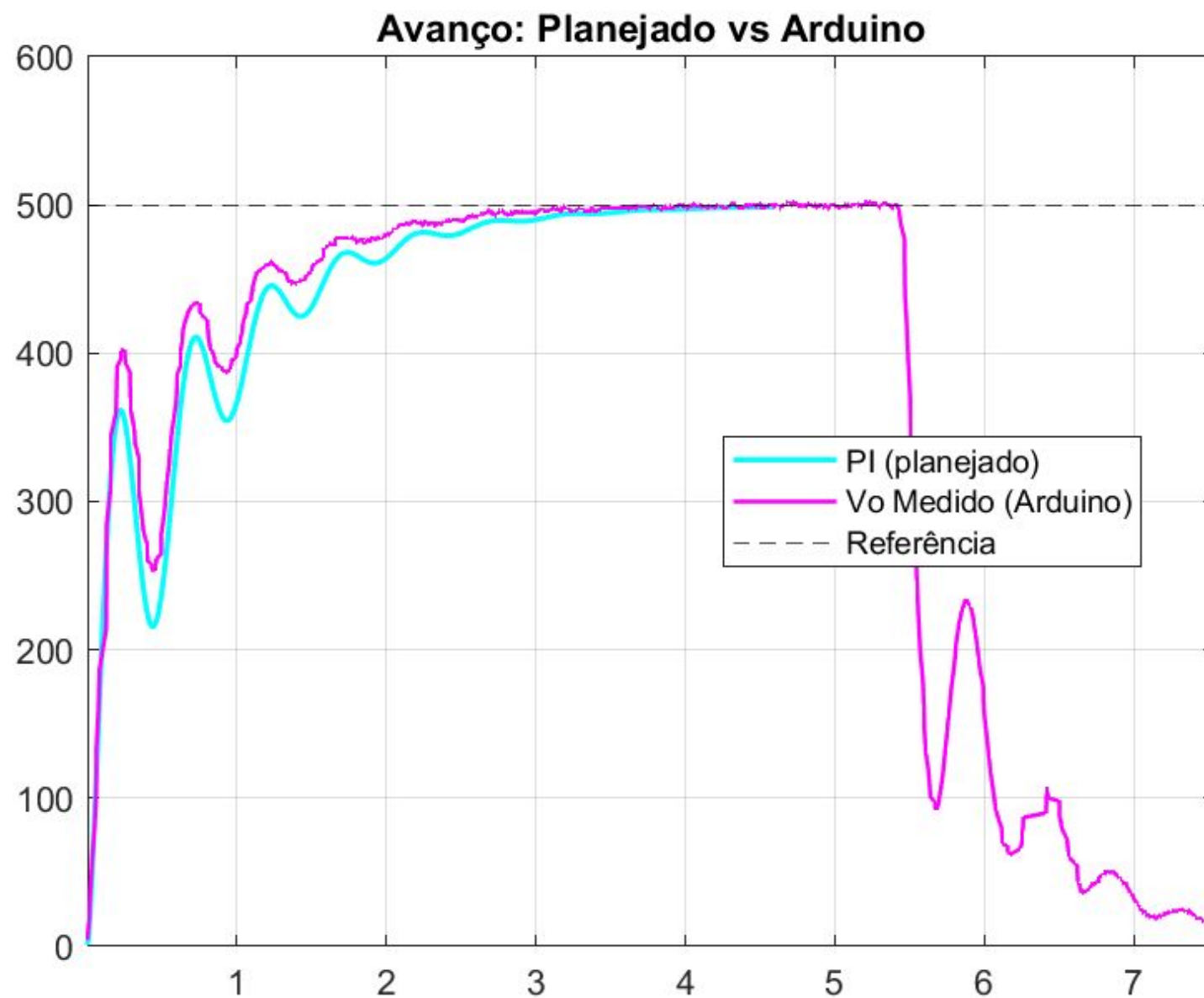
Equação de diferenças:

$u[k] = b0 \cdot e[k] + b1 \cdot e[k-1] + b2 \cdot e[k-2] - a1 \cdot u[k-1] - a2 \cdot u[k-2]$



Avanço + PI — Análises





7

Controlador PID (Robusto)

Desempenho

Sobressinal

~0 %

Tempo acomod.

~3 s

Kp_pid = 0.25; Ki_pid = 2.25;

Kd_pid = 0.025;

Kp_pid = 0.25; Ki_pid = 2.25; Kd_pid = 0.025;

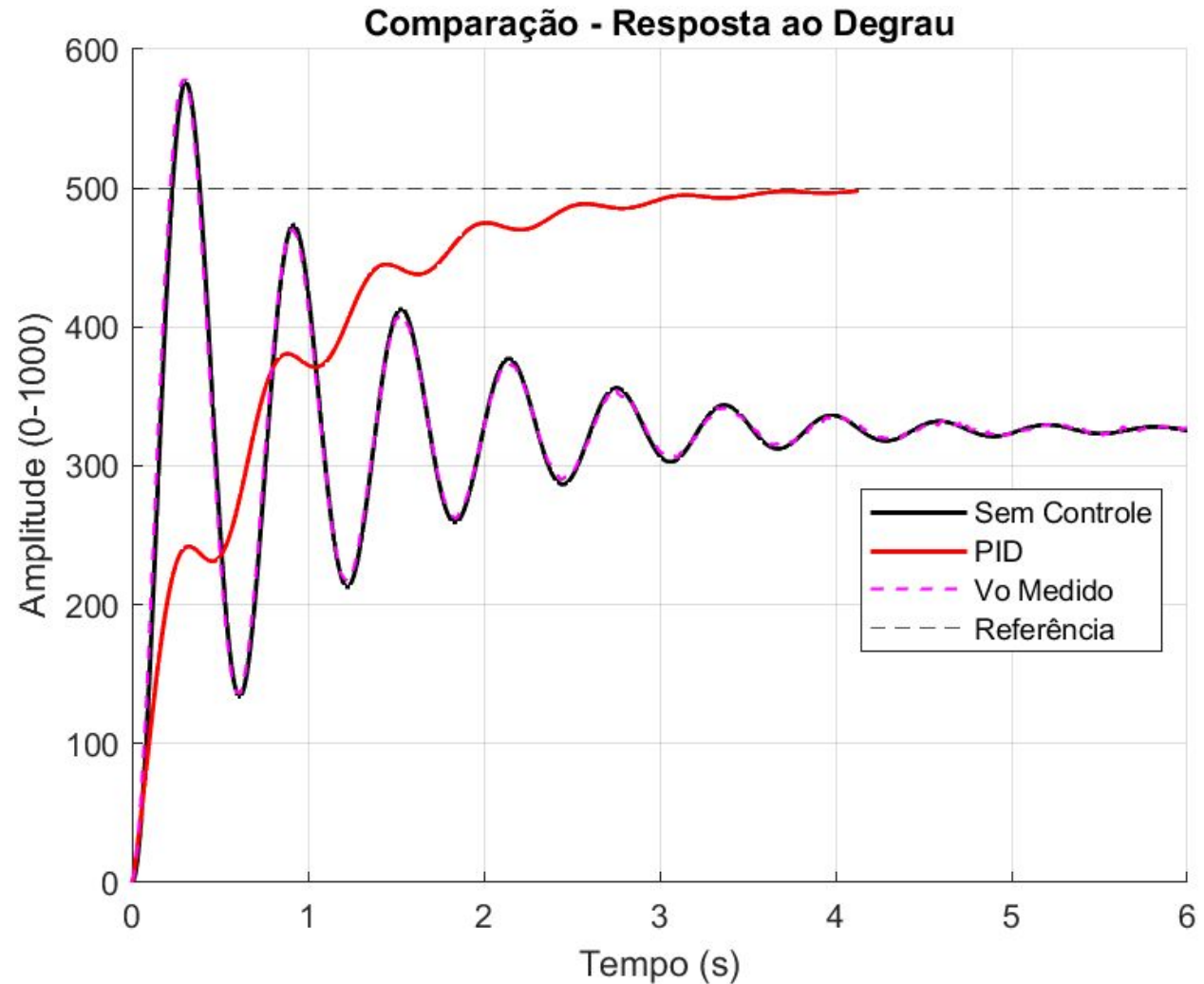
Cp = tf(Kp_pid, 1, T);

Ci = tf(Ki_pid*T, [1 -1], T);

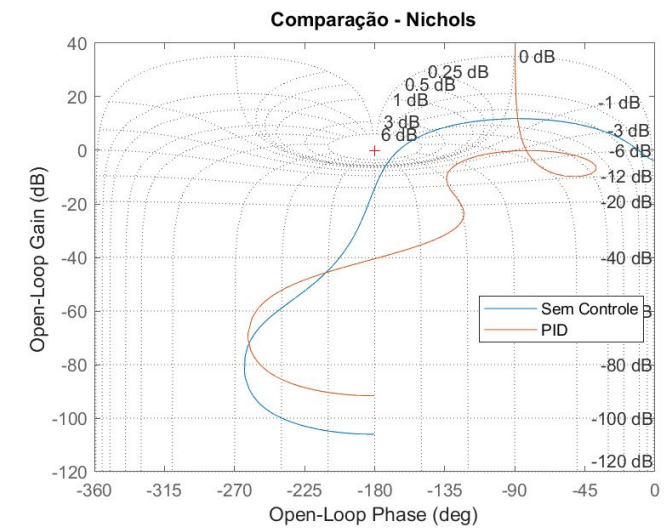
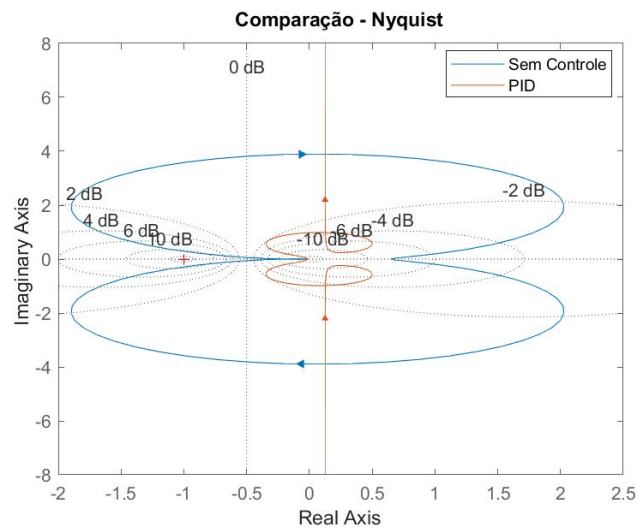
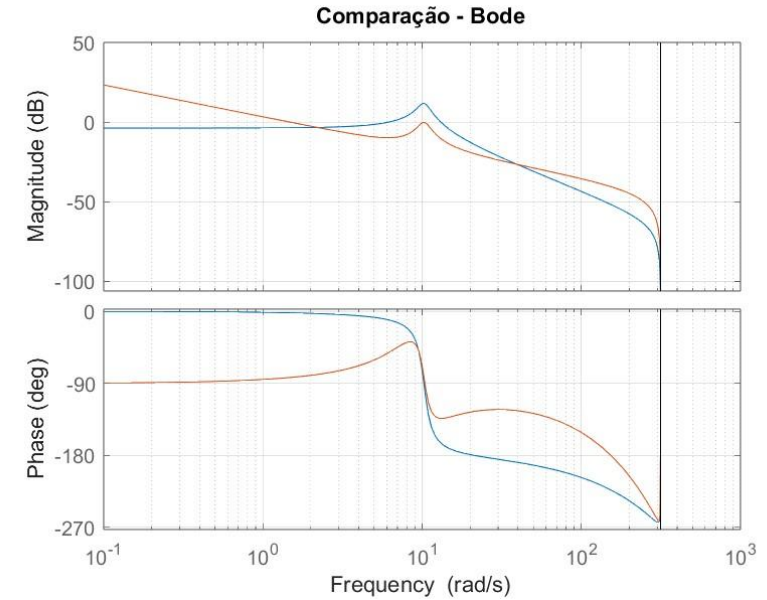
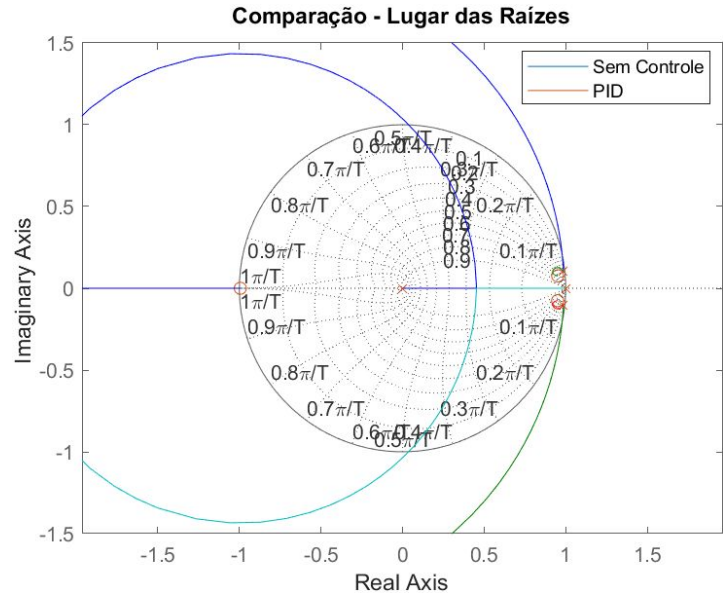
Cd = tf(Kd_pid/T*[1 -1], [1 0], T);

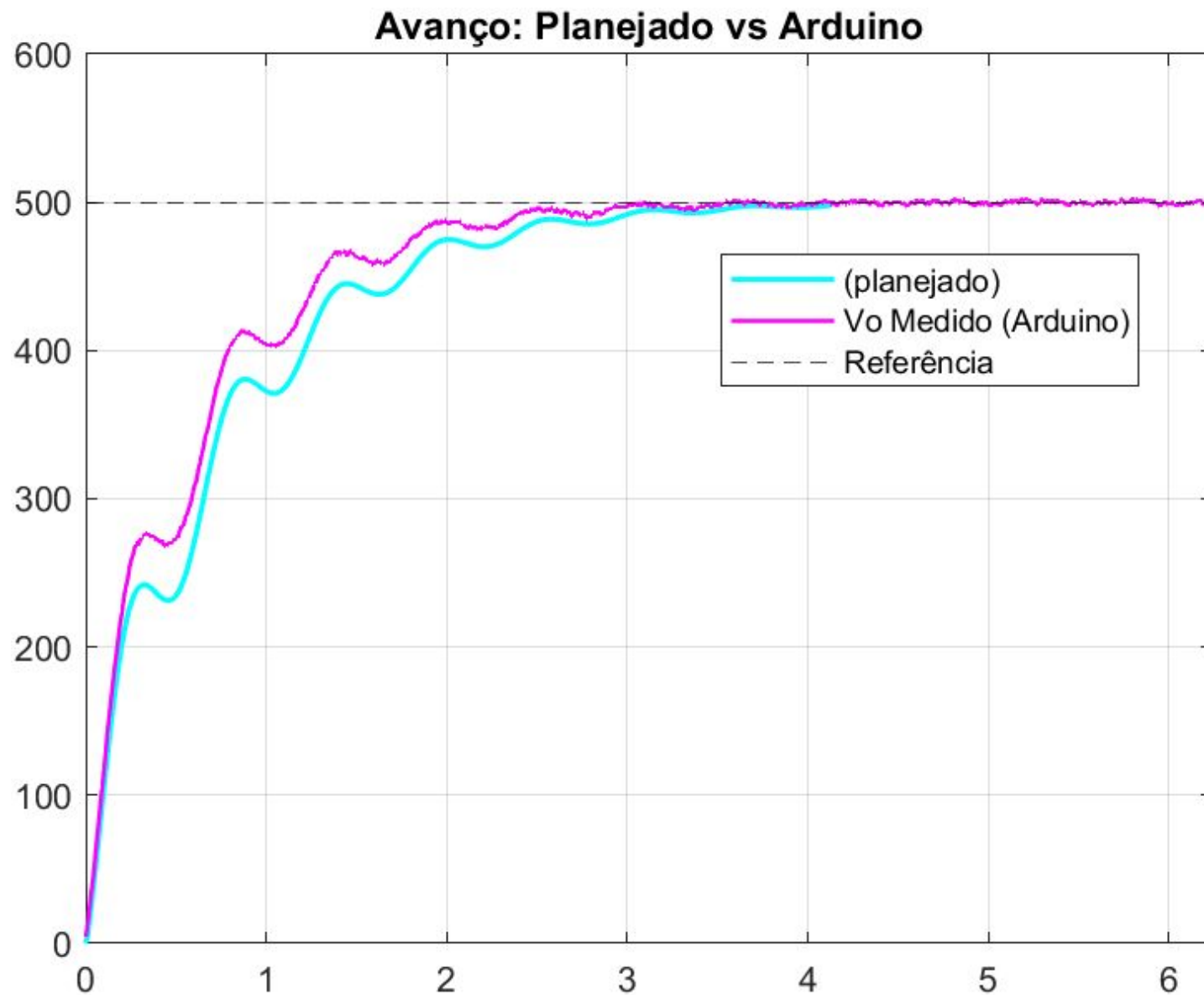
C_PID = Cp + Ci + Cd;

MF_PID = feedback(C_PID*G, 1);



★ Melhor desempenho global



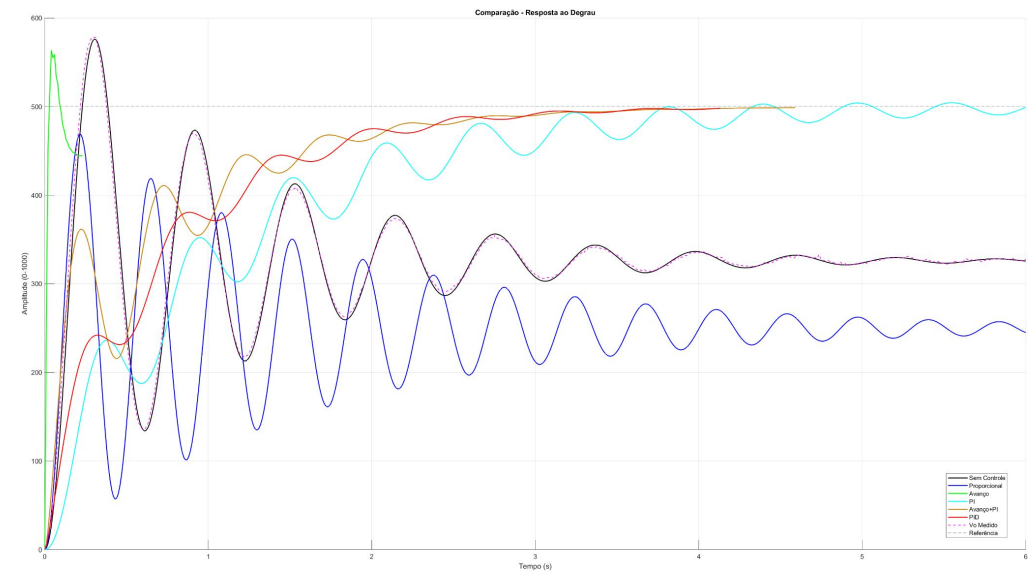




Comparação Geral dos Controladores

Controlador	RiseT (s)	SettleT (s)	OS (%)	GanhoDC	Erro (%)	MG (dB)	MF (°)
Sem Controle	0.11	4.35	76.7	0.652	34.8	14.01	15.36
Proporcional	0.07	6.51	87.72	0.499	50.06	10.31	9.52
Avanço	0.01	0.16	26.89	0.888	11.2	11.67	58.68
PI	1.91	5.34	0.86	1.000	≈ 0	6.17	96.05
Avanço+PI	1.56	3.00	0	1.000	≈ 0	33.87	39.34
PID	1.72	2.96	0	1.000	≈ 0	40.5	97.32

As métricas avaliadas são: **RiseT (tempo de subida)**, o tempo que a saída leva para sair de uma fração baixa até próximo do valor final; **SettleT (tempo de acomodação)**, o tempo até a saída estabilizar dentro de uma faixa (tipicamente $\pm 2\%$) em torno do valor final; **OS (sobressinal)**, o quanto a resposta ultrapassa o valor final, em porcentagem; **GanhoDC**, o ganho em regime permanente, onde 1 indica rastreamento perfeito; **Erro (%)**, o erro de regime ao degrau, calculado como $(1 - \text{GanhoDC})$; **MG (margem de ganho, em dB)** e **MF (margem de fase, em graus)**, que medem a robustez do sistema — quanto maiores, mais distante da instabilidade.





Referências

**AGUIRRE, Luis Antonio. Controle de Sistemas Amostrados.
[S. l.]: [s. n.], 2020.**

**[PROFESSOR]. Material da disciplina Controle II. Disponível em:
<https://oshwlab.com/daniel_bezerra/sallen-key-duino>. Acesso em: 28 jun. 2026.**

**[SEU NOME]. Sallen-Key-Duino: códigos do projeto de Controle II.
GitHub, 2026. Disponível em: <<https://github.com/luanjjbr/Sallen-Key>>. Acesso em: 28 jun. 2026.**